

Firesong-Layered Interactive Sound Design Shooter Game-Production Report

Producer: Yin Zicheng

Student ID: 11231126

Major: Art and Technology

I. Game Overview

Firesong is a 2D side-scrolling shooter game developed with the Unity engine, based on the open-source project JetFire for secondary development and functional expansion. The game features layered interactive sound design as its core characteristic, combining rich visual and auditory experiences to create an immersive shooting atmosphere.

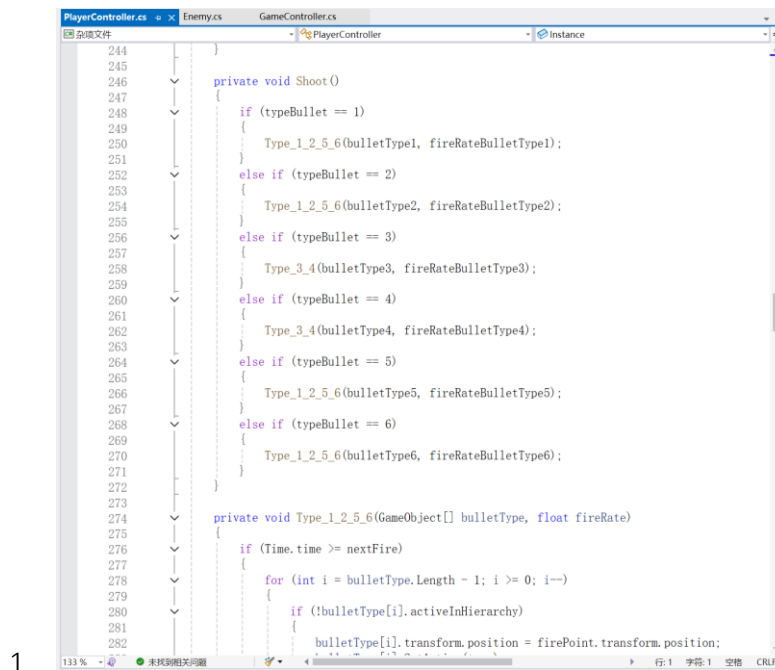
II. Pre-Production Planning

	A	B	C	D	E	F	G
	I. Character HP System						
1	System						
2	Basic HP	Max = 100					
3	HP Modifiers:						
4	Normal Attack	-10					
5	Collision/Trap	-20					
6	Bullet Attack	-30					
7	Healing Mechanic	20 Trigger: Collect "heal pack" dropped after defeating enemies					
8							
	II. Enemies & Spawn Rules						
9	Rules						
10	Enemy Types:						
11	i. Flying enemy						
12	ii. Ground Chicken	requires animation					
13	iii. Standing Lad	humanoid enemy					
14	Spawn Mechanics:						
15	Appearance	Randomly spawns 2-5 units					
16	HP	All enemies start with 100 HP					
17	Asset Requirement:	Prepare at least 3×7 sets of enemy assets					
18							
	III. Background & Scene Assets						
19	Scene Assets						
20	Background Layers:	far-ground	mid-ground	near-ground			
21		paired with:	3 sets of ground assets				
22			3 sets of sky assets				
23	Scene Element Parameters:	Prop 1	1.5f				
24		Prop 2	1.3f				
25		Prop 3	3f				
26	Asset Reward:	Complete background asset setup to unlock +8 extra assets					
27							
	IV. Bullet System						
28	Bullet Types						
29		2-hit type	4 variants				
30		1-hit type	2 variants				
31							

图表 1Game Design 游戏策划

In the early stages of game design, I established the following core elements:

1. **Game Mechanics:** Players control a character that moves vertically and shoots in side-scrolling scenes, dodging enemy attacks and traps, defeating various enemies, and collecting currency.
2. **Game Balance Design:** Including player health points, bullet damage values, and enemy behavior patterns.
3. **Asset Planning:** Organized requirements for characters, enemies, backgrounds, UI, sound effects, music, and other resources.
4. **Bullet System:** Designed six bullet types that can be randomly obtained during gameplay, each with unique visual effects and sound feedback.
5. **Enemy Design:** A total of five enemy types were designed, each with different attack patterns and behavioral logic.

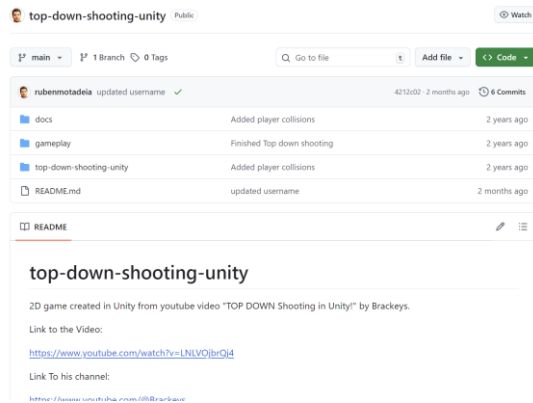


图表 2Bullet-related Code 子弹相关代码

III. Production Process

1. Initial Setup

Refactored the JetFire open-source project in Unity, developing core gameplay logic with C# scripts to implement fundamental features including player control, shooting system, enemy AI, and collision detection.

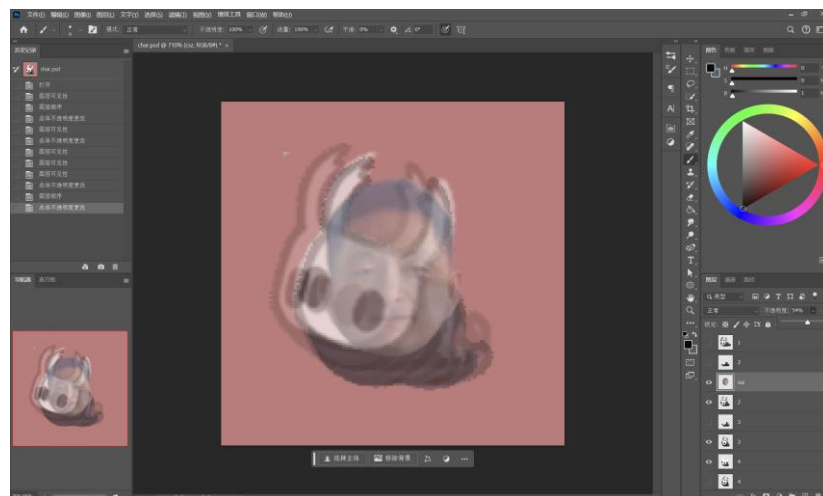


图表 3 Open-source code from GitHub 开源代码来自 GitHub

2. Art Asset Design

Hand-drawn and produced all visual assets, including:

- Character animations (walking, shooting, hit, death)
- Five enemy models and animations
- Scene backgrounds and obstacles
- Game UI interface (health bar, bullet count, score, etc.)



图表 4 Create assets with Photoshop 使用 Photoshop 绘制素材

3. Sound Design

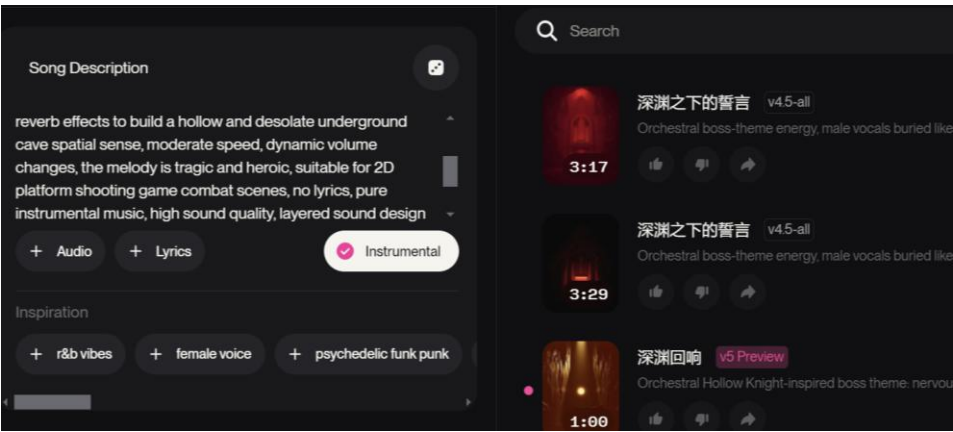
Sound design is the core feature of *Firesong*. I adopted a "layered interactive sound effect" design philosophy, deconstructing single interactive actions (such as shooting, hitting, exploding) into multiple independent audio layers. For example, a jump consists of a character shout, jetpack startup sound, and continuous operation sound; an attack

consists of a random character shout, bullet flight wind sound, and possibly hit feedback or enemy death sound. Through dynamic mixing, the layering and realism of sound effects are enhanced.

Sound Effects Required:						需要音效:					
1.Player Running						1. 玩家跑步					
2.Player Jumping						2. 玩家跳跃					
A. Jump transient						A. 跳跃瞬态					
B. Sustained flight						B. 持续飞行					
3.Player Shooting						3. 玩家射击					
A. Gunfire sound						A. 枪械声					
B. Random character sound effects						B. 随机角色音效					
C. Bullet sound						C. 子弹声					
D. Hit sound effect trigger (Y/N)						D. 是否触发命中音效					
E. Kill sound effect trigger (Y/N)						E. 是否触发击杀音效					
4.Player Damaged						4. 玩家受伤					
A. Random hit sound effects						A. 随机受击音效					
B. Bullet impact/trap trigger sound						B. 子弹打击声/陷阱触发音效					
C. Low HP warning beep trigger (Y/N)						C. 是否触发血量警告音效					
D. Death sound effect trigger (Y/N)						D. 是否触发死亡音效					
5.Enemy Sound Effects						5. 敌人音效					
A. Enemy bullet						A. 敌人子弹					
B. Enemy damaged						B. 敌人受伤					
C. Enemy running						C. 敌人跑动					
D. Enemy flying						D. 敌人飞行					
E. Enemy crawling						E. 敌人爬动					
F. Enemy death						F. 敌人死亡					
6.Collectible Items						6. 可拾取物品					
A. Random bullet pickup						A. 随机子弹拾取					
B. Health pack pickup						B. 血包拾取					
Music Required:						需要音乐:					
1.In-combat music:						1. 战斗内音乐:					
Music with mild dynamics, verse-chorus loop						动态较小的激烈音乐, 主副歌循环					
2.Out-of-combat music:						2. 战斗外音乐:					
A. Played on the initial menu						A. 初始界面播放					
B. Played on death results screen (pitch-shifted version of Music A)						B. 死亡结算界面播放, 使用音乐A的变速					

图表 5 Sound Overview 声音总览

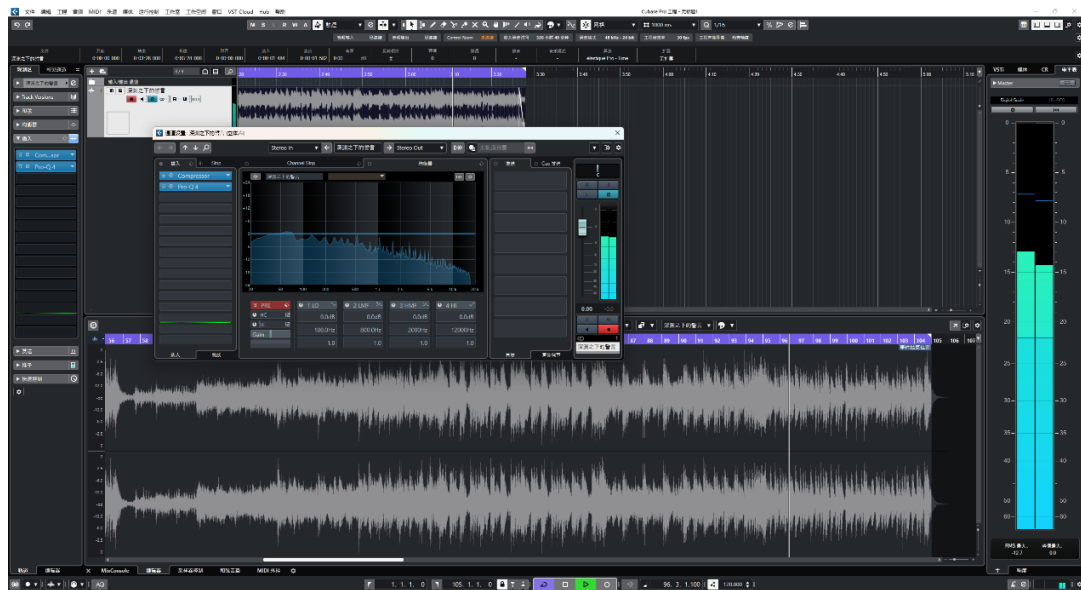
For music production, I used the prompt: *"reverb effects to build a hollow and desolate underground cave spatial sense, moderate speed, dynamic volume changes, the melody is tragic and heroic, suitable for 2D platform shooting game combat scenes, no lyrics, pure instrumental music, high sound quality, layered sound design"* to generate two background tracks in Suno AI, customized for combat and non-combat scenarios respectively.



图表 6 Music composed via Suno AI 使用 suno 生成音乐

After generation, I post-processed the music in Cubase, including tempo adjustment, key conversion, and effects processing to ensure the music matched the game's rhythm and emotional tone. The music on the game over screen is a slowed -

down version of the main menu music, conveying the sense of regret and negativity associated with player death.



图表 7 BGM processed in Cubase 使用 Cubase 处理背景音乐

For sound effect adaptation, I designed unique sound combinations for each of the six bullet types and five enemy types, and controlled trigger timing, volume, pitch, reverb, and other parameters through scripts in Unity, further synchronizing sound effects with gameplay actions.

During debugging and optimization, I repeatedly tested the trigger logic of shooting sound effects, adjusted volume balance, and fixed experience issues caused by overlapping or delayed sound effects, ensuring the immediacy and comfort of auditory feedback.

4. Feature Refinement

After two rounds of playtesting, bugs in the game logic were fixed, and game difficulty and pacing were adjusted.

IV. Conclusion

As a course project integrating art and technology, *Firesong* strives for innovation not only in visuals and gameplay but also in sound design. Through layered sound design and a dynamic audio system, I achieved a highly synchronized audiovisual interactive experience, delivering a more immersive shooting game experience for players.